29. THE RENAL SYSTEM : GROWTH AND DIFFERENTIATION

DURATION : 05:58 STUDY SHEET

COURSE SUMMARY

This video deeply explores the growth and differentiation of the renal system during embryogenesis. It highlights the central role of the liver as a major organiser, whose dynamics and growth influence the development of surrounding structures. By analysing the process of renal 'ascent', which is actually the result of the expansion of the rest of the body, as well as the phenomenon of separation between the bladder and the rectum, this session offers valuable insights into embryological reference points. Narrations on renal motility and retroperitoneal articulation demonstrate how cooperation between ectoderm and endoderm, under the influence of the mesoderm, is essential for the harmonious formation of the urinary organs.

THE RENAL SYSTEM: GROWTH AND DIFFERENTIATION

THE DYNAMIC OF POSTERIOR GROWTH AND ITS ROLE

The **posterior** part of the embryo experiences much faster **growth**, creating a kind of **"tearing"** in the lower intestine. This separation between the lower intestine and the allantoic process changes the dynamics of the posterior part.

This phenomenon is caused by a **differential growth rate** of the posterior part compared to the anterior part, maintained by the allantois and organised by the liver.

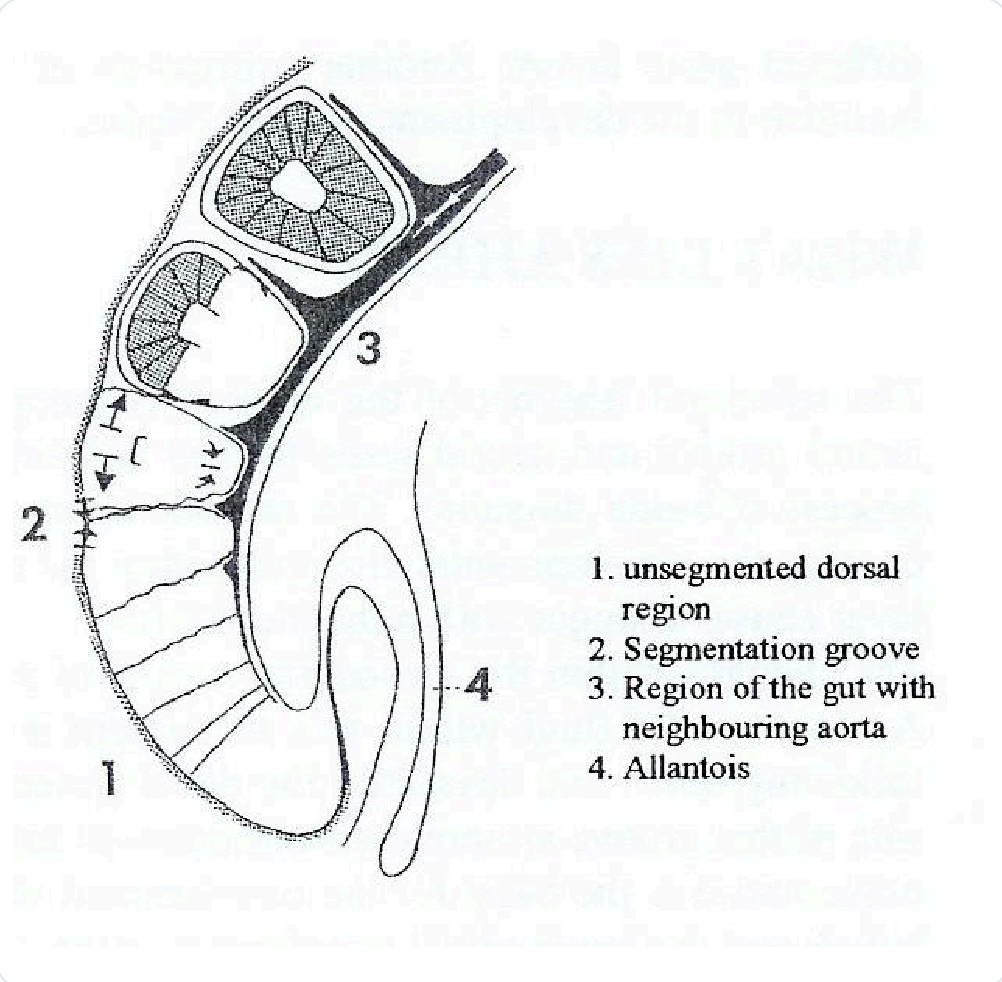


FIG. — Embryo sagittal section

THE LIVER: SYSTEM DRIVER AND ORGANISER

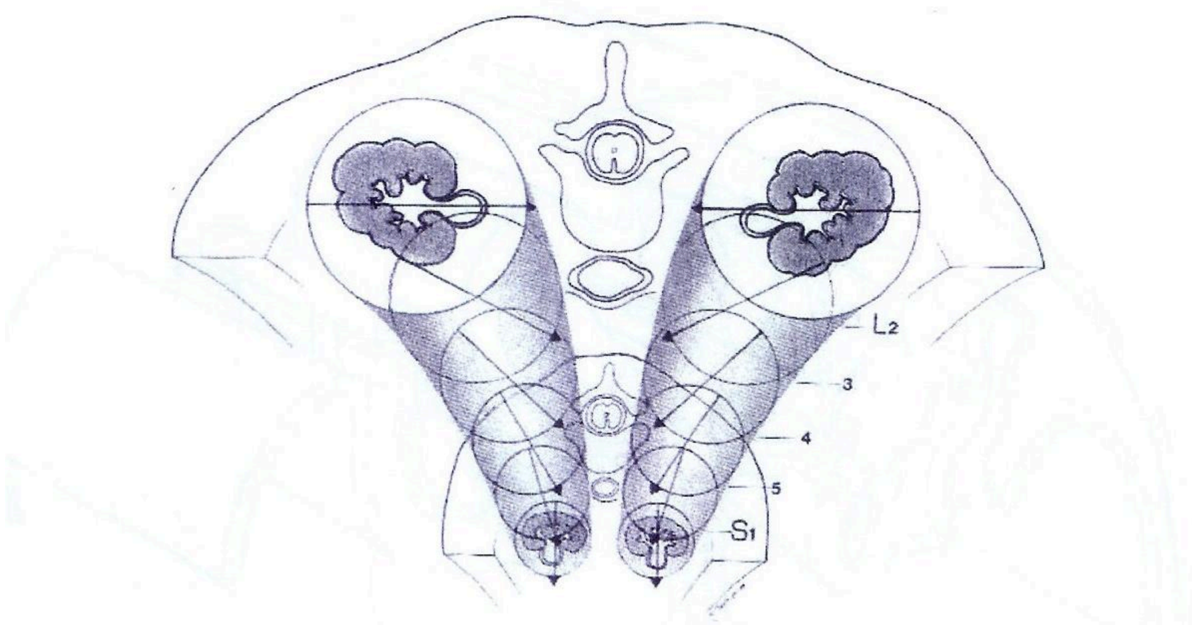
It is crucial to understand this dynamic, this **diaphragmatic** relationship, and this overall growth. The liver, through its dynamics, becomes a **major organiser**.

As an enormous organ, the liver plays an essential role in the development of the entire system through its growth and activity.

THE POSITIONING OF THE KIDNEYS AND ADRENAL GLANDS

The kidneys appear to **"ascend"** from their sacral origin to the lumbar region, but this **"ascension"** is an illusion. The adrenal glands, induced by the tilting of the liver, are initially enormous and descend with it. They give the impression of regressing, but it is actually the environment that grows around them.

It is important to think about embryonic development in terms of **changing references**. The growth of posterior structures positions the kidneys. It is not that they ascend, but that the rest descends and grows.



ıchéma montre le repositionnement des métanéphros entre la 6ème et la 9ème semaine. La croissance décrite en colonnes. Les flèches indiquent le changement progressif du pelvis rénal de ventral en médi-
Le repositionnement est présenté par rapport à la position de la colonne vertébrale et de l'aorte.

FIG. — Embryonic renal ascent

THE ROLE OF THE LIVER AND THE TRANSVERSE SEPTUM

The liver creates a **compression** from the upper to the lower part, essential for evolution. It is connected to the **transverse septum**, which, although it ascends high (embryonic origin at C3), does not move. It is the entire surrounding system that grows, giving the appearance of a **descent**.

The transverse septum, which does not move, becomes a **fulcrum** and finishes its descent at L2-L3. In reality, it is the entire environment that has moved.

THE DIFFERENTIATION OF THE URINARY TRACT

The **bladder** differentiates from the endoderm of the hindgut. The posterior growth rate of the neural tube is such that it **"tears"** the space between the future rectum and the future bladder, provided by the allantois.

Initially, the bladder and rectum formed a single structure. This separation of the cloacal space from the rectum establishes the different structures. The kidneys (metanephros) remain in place and become a **fulcrum** in relation to the posterior part of the pelvis.

THE RETROPERITONEAL ARTICULATION

The path at the kidney level reveals a **longitudinal axis** of the embryo. At the sacrum, there is a fulcrum, then the renal lodge.

The essential work is the opening of **motility** at the kidney level. The anterior compression explains their shape.

The retroperitoneal region becomes an **articulation** created by two movements:

1. The longitudinal growth of the neural tube.

2. The endodermal movement.

The integrity of these two functions depends on this intermediate articulation, which is an articulation between the **ectoderm** and the **endoderm**. The parietal system is dependent on the **mesodermal** system, which makes the mesoderm a royal pathway in embryonic development.

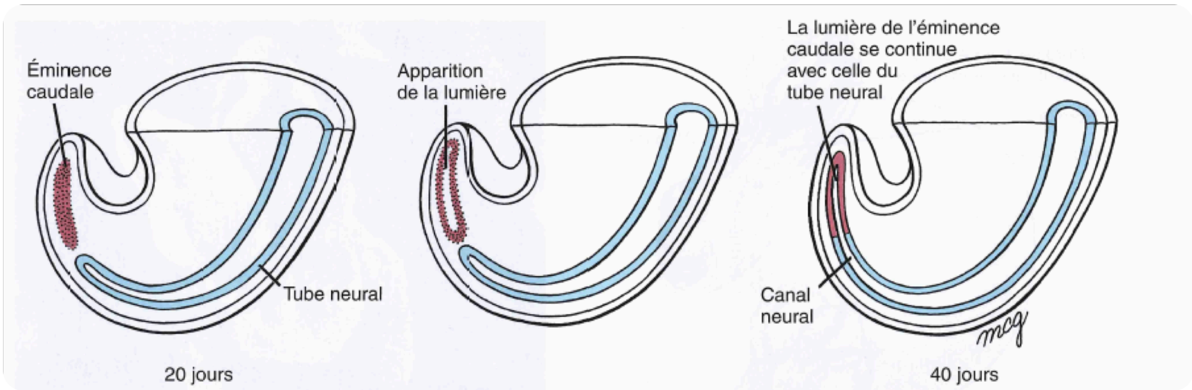


FIG. — Eminence/neural canal development